CAI 4105 · CHALLENGE 1 · REGRESSION

Apartment Rental Price Prediction

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Carlos

The problem and the data

7,398
training listings with known rent

2,467
test listings we had to price

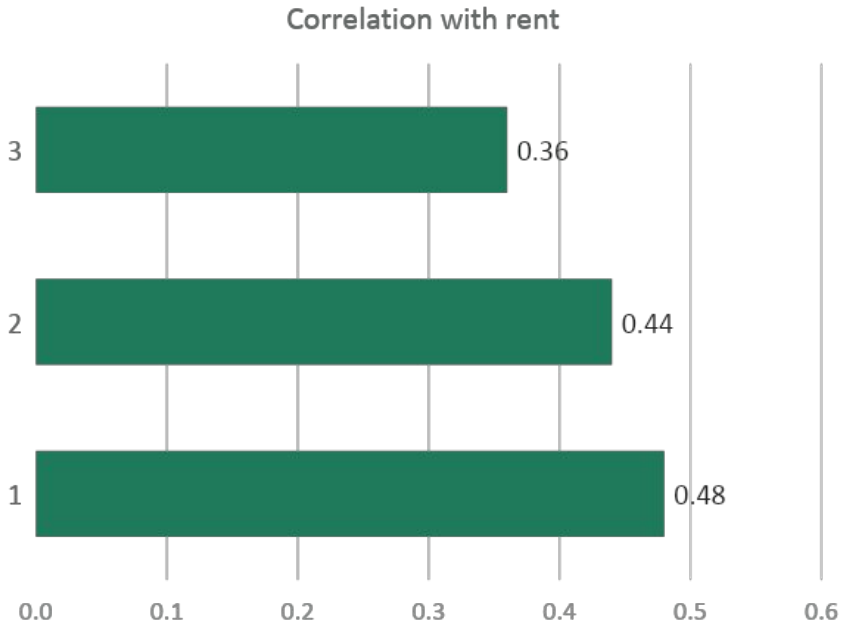
8
predictive features

Goal: predict monthly rent in dollars

Features: bedrooms, bathrooms, square feet, latitude, longitude, state, has_photo, category. We score with RMSE: "on average, we're off by $X a month."

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What the data told us



Rent is right-skewed

Median $1,274 · mean $1,463 · max $8,800

Location matters a lot

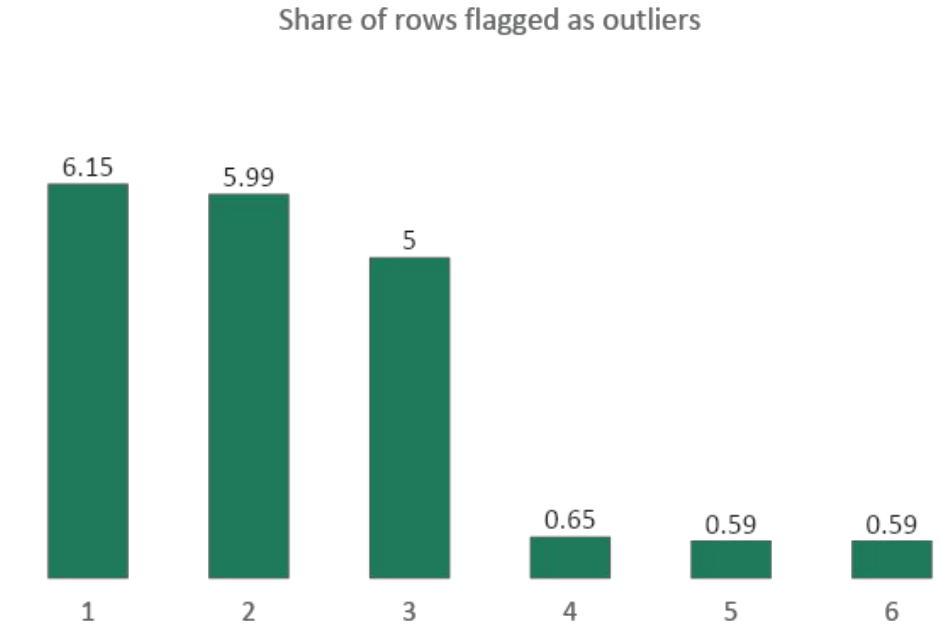
Median rent: HI $2,850 · CA $2,325 · MA $2,274

Clean, with two quirks

No missing values · category is 99.96% "apartment" · WY only in test

Carlos

Outlier investigation (1.5 × IQR)



Decision: keep every row

455 sq ft, 443 price and 370 bedroom rows flagged. They look like real listings, so none were removed.

Handle skew another way

Log-transform the target and use tree models, which extreme values don't throw off.

Carlos → Abhiram

Preprocessing and validation

1

Drop

ID and the near-constant category

2

Encode

One-hot state and has_photo; unseen WY ignored

3

Scale

Standardize numbers, for linear models only

4

Validate

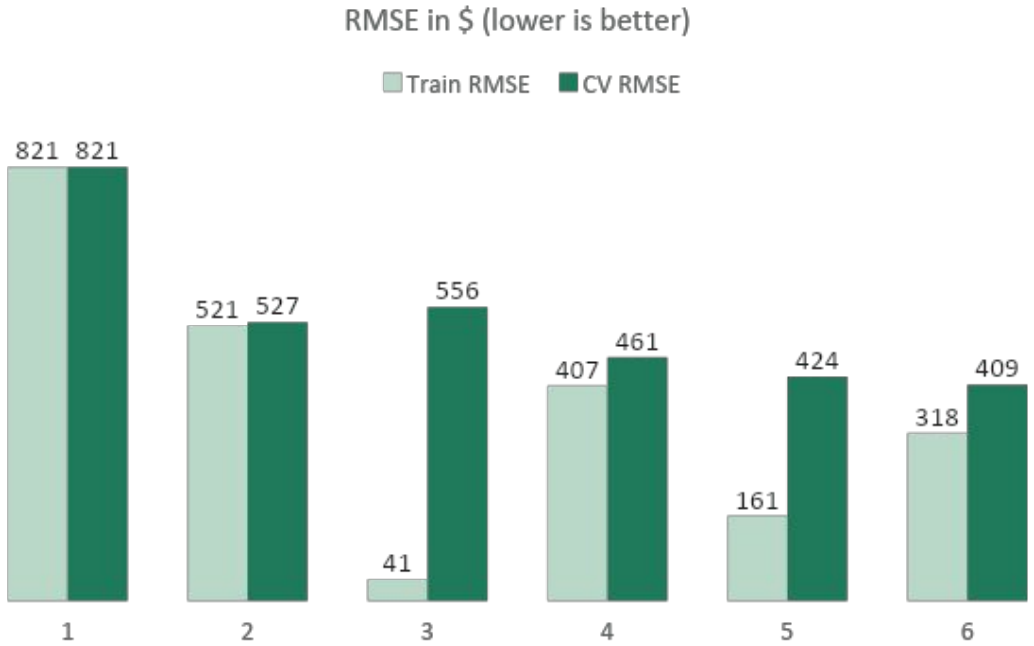
5-fold shuffled CV, seed 42, train data only

No data leakage

Every step sits inside a scikit-learn Pipeline, so encoders and scalers only learn from training folds. The test set is used once, at the very end.

Abhiram

Four required models, compared fairly



Decision Tree overfits

$41 train vs $556 CV

Linear underfits

About $520 on both

Ensembles lead

Already at $409–$424

Abhiram

Better features, log target, Grid Search

4 new features

log_sqft · total_rooms · sqft_per_bed · bath_per_bed

Log target

Train on log(price), convert back; error still in $

GridSearchCV

Same 5 folds; GB best: learning rate 0.08, depth 5

Tuned model	Train	CV RMSE
Gradient Boosting	**$191**	**$390**
Hist Grad. Boosting	$262	$396
Random Forest	$189	$421
Decision Tree	$345	$487
Ridge	$507	$512

Abhiram

Tuned Gradient Boosting wins

$390

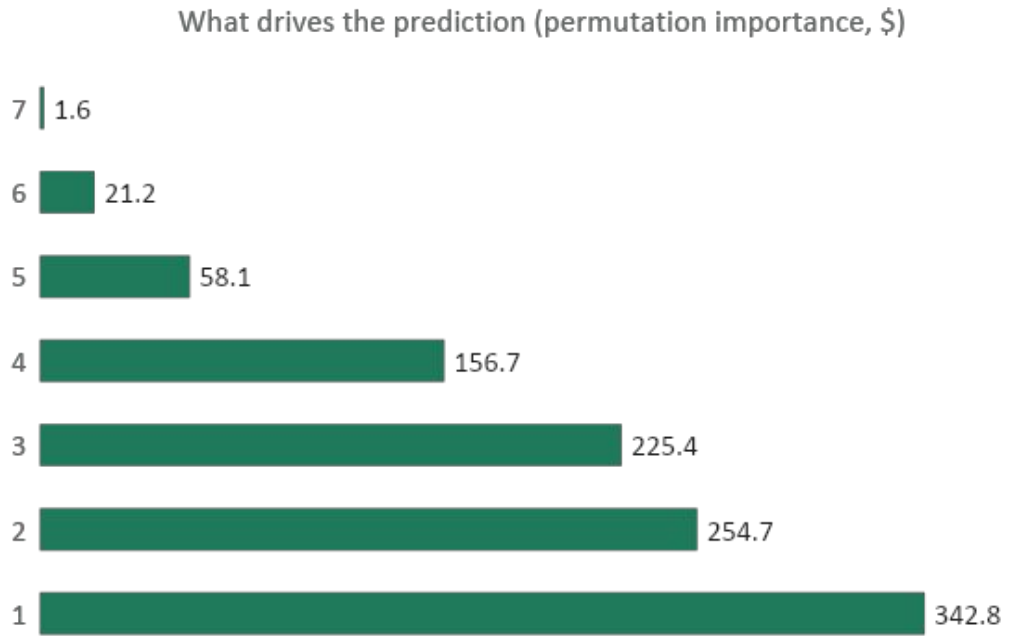
average error (5-fold CV)

−52%

vs. the $821 mean guess

2,467

test rents predicted



KEY TAKEAWAYS

01 Location and size drive rent; straight lines can't capture it

02 Pipelines + 5-fold CV kept every comparison fair and leak-free

03 Tuned Gradient Boosting cut error from $821 to about $390

Thank you. Questions?